57% of source accounts appear two-year dormant — recovery must start with data validation, not outreach

WHERE THE DORMANT OPPORTUNITY SITS

2,000
source accounts in 301 root groups

1,144
provisionally two-year inactive — 57.2% of source

295
root groups contain the 1,144 modeled accounts

Composition of the 2,000 source accounts by inactivity evidence



$201.5k
analytical expected net program NPV — base hazard, 15% hypothetical offer, 12 months, 1,144 accounts. Illustrative unit economics; not verified profit or causal uplift.

0
accounts are authorized for customer contact today. Every route is a planning candidate pending eligibility, ownership and permission checks.

Recommendation: evidence-gated program

1. **Validate**
   Confirm fiscal-year inactivity, partner ownership, health and service evidence before any account is targeted.
2. **Prioritize**
   Rank opportunities with documented, versioned assumptions and keep evidence gaps visible beside every score.
3. **Route**
   Assign partner-led, direct-enterprise or digital-nurture planning workflows; hold unresolved accounts for review.
4. **Pilot & measure**
   Run controlled pilots with an untreated comparison group and track observed outcomes before any ROI claim.

Decision: fund a data-validation sprint and a measured pilot — not a full-scale reactivation campaign.

Source: Case_Study_1_Install_Base_Reactivation.csv (2,000 rows; SHA-256 6863ea00…a229bc); frozen Phase 1–4 outputs. “Provisional” = source-reported years_inactive ≥ 2; fiscal-year eligibility unconfirmed. The brief’s “>50% of install base, +21% CAGR” cannot be verified from this extract. Historical spend is not ARR; NPV is illustrative, not causal.

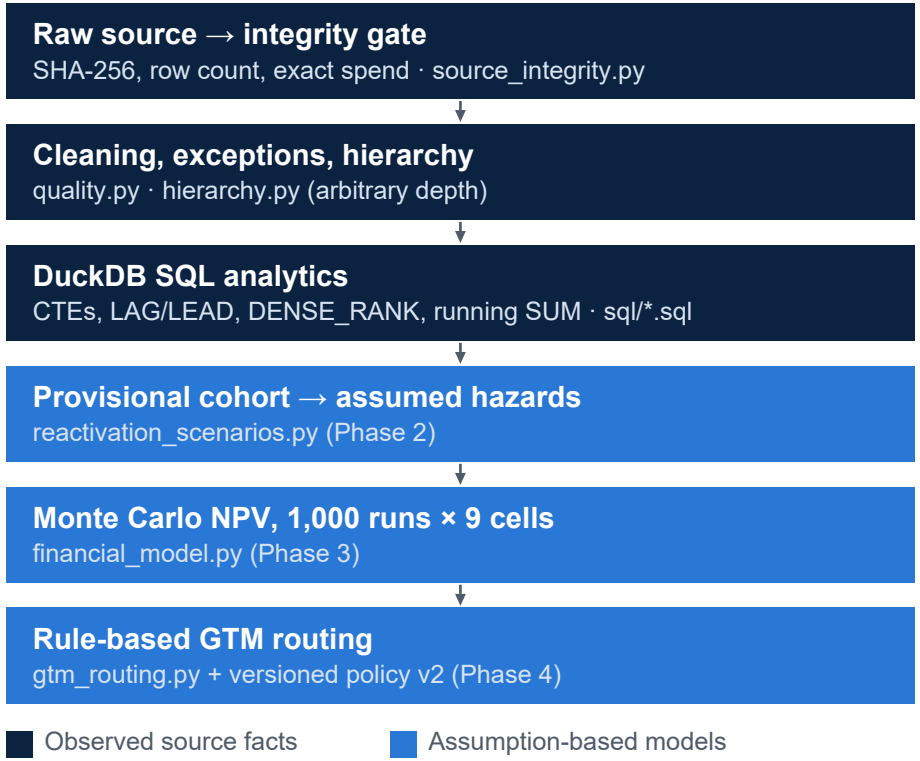
Systematic hygiene logged 4,889 data-quality exceptions and keeps observed facts separate from modeled assumptions

INJECTED DEFECTS AND HOW EACH WAS HANDLED

Defect named in the brief	Observed in the 2,000-row extract	Handling
Duplicate accounts across parent-child	0 duplicate IDs; duplicate, cycle and self-reference checks retained	Flag for review; never auto-merge
Region string formatting	6 raw variants → 5 regions; 667 France rows normalized	Trim + casefold mapping; raw value kept
Missing SFDC health score	844 unknown (444 'nan', 400 'Missing')	Kept unknown and flagged; no imputation
NaN partner-managed flag	654 unresolved	Kept unknown; routes to hold (R3)
Sentinel 999 open tickets	322 sentinels vs. 324 genuine zeros	Excluded from valid ticket counts
Inactivity definition (found, not injected)	1,144 agree · 562 conflict · 294 unknown	Provisional cohort; conflicts held

4,889 exception events across 1,968 accounts · hierarchy resolved: 301 roots, 1,699 children, 0 unresolved

MODELING ARCHITECTURE (IMPLEMENTED MODULES)



Why there is no observed decay curve: booking dates rise exactly with account ID and every contract starts exactly 365 days earlier, so LAG/LEAD gaps reduce to ID arithmetic. Revenue-decay velocity and historical reactivation timing cannot be estimated from this extract; the decay SQL (LAG, LEAD, DENSE_RANK) is executed on a labeled synthetic fixture.

Source: Phase 1 exception register and metrics (analysis date 2026-10-09). The brief describes injected duplicates; the supplied file has zero duplicate account IDs, so detection is implemented but no merge is invented. Account identifiers and row-level outputs are confidential and excluded from every shared artifact.

Assumed hazards imply 66–206 expected first reactivations within 12 months — a planning range, not a forecast

LANE A · METHOD DEMONSTRATION

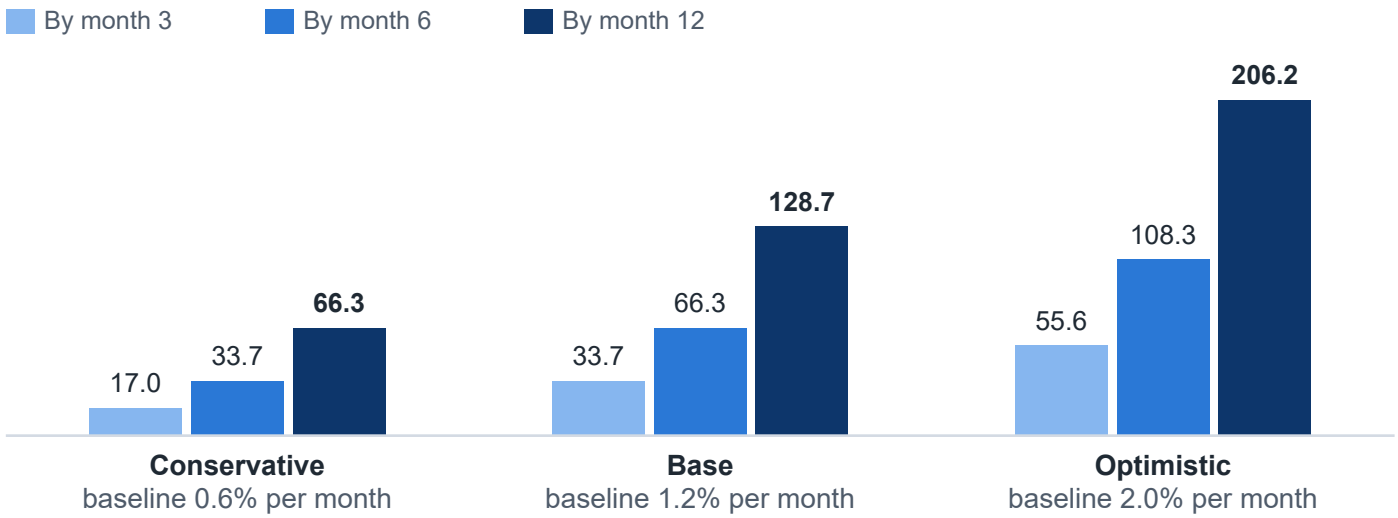
Cox proportional hazards on synthetic event histories

Synthetic covariate	True β	Fitted HR (95% CI)
Health proxy	+0.45	1.66 (1.47–1.87)
Engagement proxy	−0.35	0.70 (0.66–0.74)
Partner proxy	+0.25	1.25 (1.11–1.41)

- 1,600 invented subjects: 1,074 events, 526 right-censored.
- Concordance 0.62 in-sample and 0.60 on an independent synthetic holdout.
- Schoenfeld proportional-hazards p-values 0.28 / 0.25 / 0.94: no violation detected.
- **Shows the lifelines workflow recovers known effects. It is not an Everpure model: the extract has no customer event histories.**

LANE B · EVERPURE PLANNING SCENARIOS

Assumption-based exponential hazards for the 1,144 modeled accounts — expected cumulative first reactivations



- Counts are cumulative expectations, not observed conversions: the 3-, 6- and 12-month values must not be added. No campaign treatment effect is identified.
- P(reactivate by month t) = 1 − exp(−λ·t·$e^{x\beta}$), with assumed effects for health, partner ownership, child status and valid tickets.
- Unknown health is scored at the reference level and lifts rankings: scoring it as Yellow or Red cuts base 12-month reactivations from 128.7 to 121.1 or 112.4.

Source: frozen Phase 2 outputs (reactivation_scenario_summary.csv, scenario_sensitivity.csv, synthetic_cox_coefficients.csv); assumption register v1, fingerprint 6195ba2e…. Probabilities are ASSUMPTION_BASED_NOT_CALIBRATED; the Cox fit uses invented subjects only.

Illustrative economics are positive in the base case (~$202k at a 15% offer) but depend heavily on unverified assumptions

ANALYTICAL EXPECTED NET PROGRAM NPV — BASE HAZARD, BY HYPOTHETICAL OFFER



MONTE CARLO DISTRIBUTION — BASE HAZARD, 15% OFFER, 1,000 RUNS

Sample mean	Median	5th percentile	95th percentile	Runs with NPV < $0
$202.9k	**$202.6k**	**$150.7k**	**$256.5k**	**0 of 1,000**

Analytical expectation $201.5k (closed form) vs. simulation sample mean $202.9k: a +1.38 standard-error difference; the largest across all nine cells is 1.88.

How to read the result

- **5% scores highest only because the model holds reactivation hazard fixed across offers.** 15% and 30% would need 1.13× and 1.41× the hazard to match it — not evidence that 5% is optimal.
- Across the nine cells the expected NPV ranges from $15.9k (conservative, 30%) to $464.6k (optimistic, 5%). The cells are alternatives; never add them.
- Committed combined adverse stress ($750 revenue, 45% margin, $150 outreach, $300 activation, 24% rate, 3 paid months) turns NPV to **−$124.6k**.
- Net program value under analyst-selected unit economics — not verified profitability or causally identified incremental uplift.

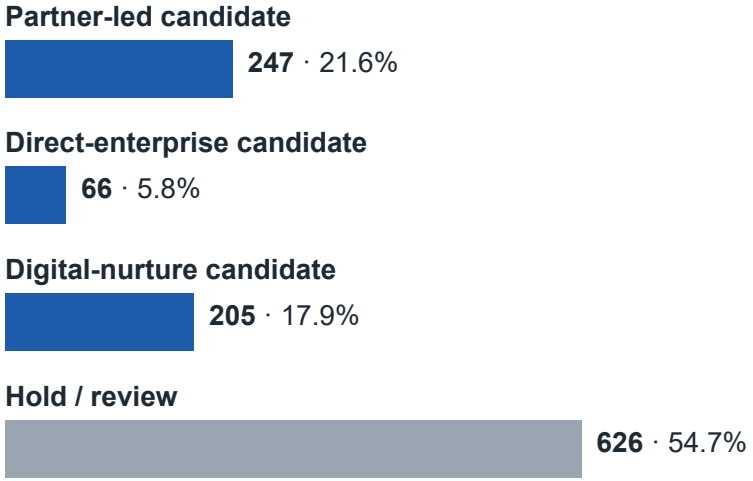
ILLUSTRATIVE UNIT ECONOMICS (ANALYST-SELECTED, NOT OBSERVED)

$1,000 gross revenue per activated account per month	**60%** contribution margin	**$100** outreach per modeled account	**$200** activation cost per reactivation	**12%** annual effective discount rate	**12 months** financial horizon, no churn

Source: frozen Phase 3 outputs (financial_scenario_matrix.csv, monte_carlo_npv_summary.csv, financial_sensitivity.csv); assumption fingerprint b953c02d…; NumPy PCG64 seed 20261009 with common random numbers. P5/P95 are conditional simulation percentiles, not confidence intervals. Label: ILLUSTRATIVE_NET_PROGRAM_NPV_NOT_CAUSALLY_ATTRIBUTED.

Route 518 accounts to conditional planning channels, hold 626 for review, and release outreach only through gated pilots

CONDITIONAL ROUTES · 1,144 MODELED ACCOUNTS



Holds: 367 unknown partner ownership (R3) and 259 needing service recovery — Red health or ≥ 5 valid open tickets (R4).

EVIDENCE QUALITY & HIERARCHY

288 · 418 · 438
High / Medium / Low illustrative priority (fixed NPV cutpoints)

187 of 288
High-priority accounts have unknown health

235 of 295 roots
carry a known partner-ownership conflict (full-source membership)

282 roots · 1,131 accts
require parent-child coordination before contact

0
authorized customer-contact actions

RELEASE GATES BEFORE ANY OUTREACH

Gate 1	**Validate** — fiscal eligibility, partner ownership, health and service barriers
Gate 2	**Coordinate** — resolve parent-child ownership; prevent overlapping campaigns
Gate 3	**Pilot** — controlled partner / direct / digital pilots where permissions exist
Gate 4	**Measure** — observed reactivation, uplift vs. control, actual cost and margin
Gate 5	**Scale** — recalibrate the model; scale only when evidence supports it

Decision: approve a data-validation sprint and a controlled, measured pilot — not a full-scale reactivation campaign. The 518 routed candidates are not approved contacts.

Full Analytical Evidence & Diagrams — Google Drive

Executable Code, SQL & Notebooks — GitHub

Source: frozen Phase 4 aggregates via phase5_input_contract.json v2 (policy fingerprint 1998cc4a…). All dispositions are PLANNING_ONLY with execution_authorized = false. Priority tiers use fixed base/15% NPV cutpoints; the modeled NPV is not attributable to the 518 routed accounts alone.

Appendix A — Reproducibility, evidence links and deliverables map

TECHNICAL ARTIFACTS

Full Analytical Evidence & Diagrams — Google Drive
Rendered diagrams, aggregate result tables, notebooks and this deck

Executable Code, SQL & Notebooks — GitHub
Sanitized package with a fresh Git history; runs without confidential data

WHAT THE CODE PACKAGE CONTAINS

Path	Contents
`notebooks/00_synthetic_reviewer_demo`	End-to-end run on invented accounts: cleaning, SQL, hazards, Monte Carlo, routing (no confidential data)
`notebooks/01–04`	Data foundation & SQL · survival · Monte Carlo NPV · GTM routing (full runs need the authorized source)
`src/everpure_case/`	Python modules: integrity, cleaning, hierarchy, survival, finance, routing
`sql/`	DuckDB SQL: CTEs, LAG/LEAD, DENSE_RANK, running windows
`tests/ · docs/ · outputs/`	pytest suite · methodology, contracts, registers · frozen aggregate results with SHA-256 manifests

Brief deliverables: executable Python/SQL notebook → code package (A, B) · three-tier Monte Carlo NPV summary → slide 4 and D · routing flowchart → slide 5 and F · five-slide deck → pages 1–5.

TWO REPRODUCTION PATHS

Path A — authorized full reproduction. Place the original assessment CSV at `data/raw/`; the integrity gate checks its SHA-256, 2,000 rows and $2,541,024,291 recorded spend. Then run the Phase 1→4 pipelines; the validators re-derive every route and aggregate.

Path B — synthetic demonstration. `pip install -r requirements.txt`, then run notebook 00 and the synthetic tests. Invented SYN_ accounts only; it shows the mechanics and never reproduces Everpure results.

PROVENANCE FINGERPRINTS (SHA-256)

Source CSV	`6863ea0014f050355bf9e0c5166e2284…`
Phase 2 assumptions	`6195ba2e90c95976ed91b7bdd4250934…`
Phase 3 financials	`b953c02d3fdcfacc62d9e6755889040e…`
Phase 4 policy	`1998cc4a68c88160f01d89ca0abfe7c2…`
Phase 5 input contract	`89bfe72319e31112fa64b837b741e664…`

Environment: Python 3.12 · pandas 2 · NumPy 2 · DuckDB 1 · lifelines 0.30 · matplotlib 3.10 · nbclient/ipykernel · pytest. Frozen baseline commit 38b1f43 (Phase 4).

Neither link is a substitute for this PDF: pages 1–5 are the executive deck and these pages carry the technical evidence. Confidential source rows, account identifiers, root mappings and account-level model outputs are excluded from every shared package.

Appendix B — SQL engineering and data hygiene (implemented code)

sql/spend_concentration.sql — CTE chain, DENSE_RANK, running SUM

```
WITH account_spend AS (
    SELECT h.root_account_id, a.account_id, a.historical_spend_usd_num AS spend_usd
    FROM accounts a JOIN hierarchy h USING (source_row_number)
    WHERE h.hierarchy_status = 'resolved'
), root_spend AS (
    SELECT root_account_id, COUNT(*) AS account_count,
           SUM(spend_usd) AS additive_recorded_spend_usd
    FROM account_spend GROUP BY root_account_id
), ranked AS (
    SELECT *, DENSE_RANK() OVER (ORDER BY additive_recorded_spend_usd DESC) AS spend_rank,
           SUM(additive_recorded_spend_usd) OVER () AS all_resolved_spend_usd,
           SUM(additive_recorded_spend_usd) OVER (
               ORDER BY additive_recorded_spend_usd DESC, root_account_id
               ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW
           ) AS cumulative_recorded_spend_usd
    FROM root_spend
)
```

sql/synthetic_account_decay.sql — LAG / LEAD / DENSE_RANK decay velocity (synthetic)

```
), sequenced AS (
    SELECT *,
           LAG(fiscal_period_index) OVER account_sequence AS previous_period_index,
           LAG(period_revenue_usd) OVER account_sequence AS previous_revenue_usd,
           LEAD(fiscal_period_index) OVER account_sequence AS next_period_index,
           LEAD(period_revenue_usd) OVER account_sequence AS next_revenue_usd,
           DENSE_RANK() OVER (
               PARTITION BY account_id ORDER BY fiscal_period_index DESC
           ) AS observed_period_recency_rank
    FROM validated
    WINDOW account_sequence AS (PARTITION BY account_id ORDER BY fiscal_period_index)
)
SELECT ..., CASE WHEN fiscal_period_index - previous_period_index = 1
                 AND previous_revenue_usd <> 0
            THEN (period_revenue_usd - previous_revenue_usd)
                 / NULLIF(previous_revenue_usd, 0) END AS consecutive_pct_change
```

EXCEPTION REGISTER (PHASE 1)

Issue (code)	Events
Reported vs. date-derived inactivity gap ≥ 6 months INACTIVITY_VALUE_GAP_6M	1,546
SFDC health missing ('nan' / 'Missing') UNKNOWN_HEALTH	844
Region whitespace / casing normalized REGION_NORMALIZED	667
Partner-managed flag missing UNRESOLVED_PARTNER	654
Reported < 2 years vs. calendar proxy ≥ 2 INACTIVITY_CONTRADICTION	562
Sentinel 999 open tickets TICKET_SENTINEL_999	322
years_inactive missing MISSING_INACTIVITY	294
Total exception events · 1,968 accounts	**4,889**

src/everpure_case/quality.py (excerpt)

```
def normalize_region(raw: str) -> str | None:
    return REGION_MAP.get(" ".join(raw.split()).casefold())

frame["ticket_sentinel"] = frame["open_support_tickets_num"].eq(999)
frame["tickets_clean"] = (
    frame["open_support_tickets_num"].mask(frame["ticket_sentinel"]))
dup = frame["account_id_key"].ne("") & \
      frame["account_id_key"].duplicated(keep=False)
```

Excerpts are verbatim from the implemented files except where marked "…" or re-wrapped; complete code is in the package. Real-data LAG/LEAD queries run on booking dates that follow the account-ID sequence, so they are reported as source-pattern diagnostics, never as decay evidence.

Appendix C — Survival methodology and assumed-hazard scenarios

Lane B — Everpure planning scenarios

- Constant monthly hazard per account: $h(t \mid x) = \lambda_s \cdot \exp(x \cdot \beta)$
- $P(T \le t \mid x) = 1 - \exp(-\lambda_s \cdot t \cdot \exp(x \cdot \beta))$; $E[N(t)] = \Sigma\, P(T \le t \mid x)$ over the 1,144 accounts
- λ_s per month: conservative 0.006 · base 0.012 · optimistic 0.020
- Assumed β: Red health −0.35; Yellow −0.15; child account −0.10; log(1 + valid tickets) −0.08; partner-managed Yes +0.15
- Unknown health, partner or tickets → reference level (0) plus explicit missingness flags

Lane A — Cox proportional hazards (synthetic)

- $h(t \mid x) = h_0(t) \cdot \exp(x \cdot \beta)$, estimated by partial likelihood (lifelines CoxPHFitter); 1,600 invented subjects, 36-month follow-up, seed 20261009
- 1,074 events, 526 right-censored; fitted log-coefficients +0.505 / −0.360 / +0.221 vs. generating +0.450 / −0.350 / +0.250
- Concordance 0.624 in-sample, 0.595 holdout; rank-based Schoenfeld tests do not reject proportional hazards
- Purpose: show a correct, diagnosable survival workflow. It is never used to score Everpure accounts.

EXPECTED CUMULATIVE FIRST REACTIVATIONS (1,144 ACCOUNTS)

Scenario	3 months	6 months	12 months	Mean P(12m)
Conservative	16.96	33.67	**66.33**	5.8%
Base	33.67	66.33	**128.72**	11.3%
Optimistic	55.55	108.35	**206.21**	18.0%

BASE-SCENARIO 12-MONTH SENSITIVITY (SAME MODEL, ONE CHANGE AT A TIME)

Case	Expected reactivations
Primary base case	128.7
Baseline hazard × 0.5	66.3
Baseline hazard × 1.5	187.4
Covariate effects × 0 / × 0.5 / × 1.5	153.4 / 140.2 / 118.8
Unknown health as Yellow / Red / observed mix	121.1 / 112.4 / 120.3
Hypothetically include 562 conflicts (n = 1,706)	191.9
…and 294 unknowns too (n = 2,000)	225.2

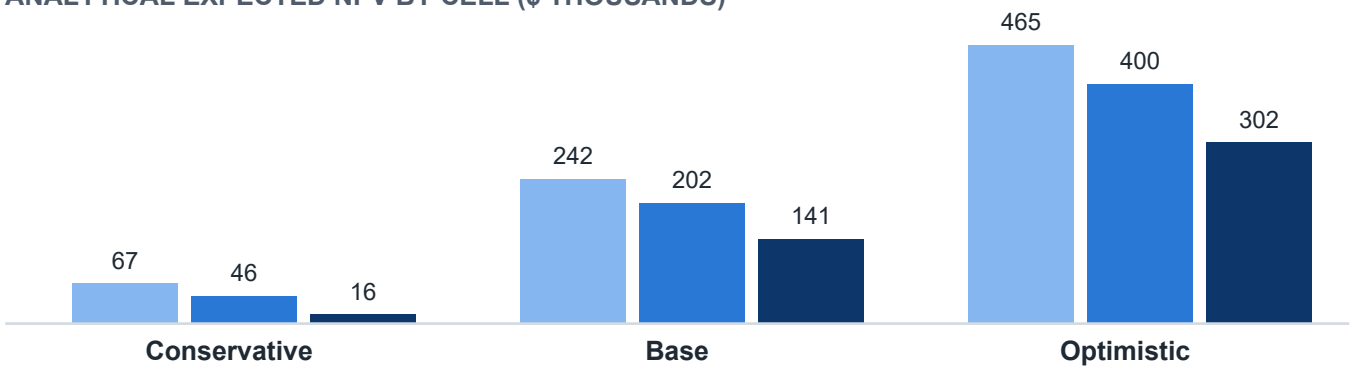
Base 12-month account probabilities range 7.6%–15.4% (mean 11.3%). Larger counts under wider populations mainly reflect larger denominators.

Lane B probabilities are scenario assumptions (register phase2-assumptions-v1), not estimates from Everpure outcomes. Lane A coefficients describe an invented experiment only. Expected counts at 3, 6 and 12 months are cumulative and never additive.

Appendix D — Monte Carlo NPV summary — all nine scenario × offer cells

Hazard scenario	Offer	Analytical expected NPV	MC sample mean	MC std. error	z	MC median	MC 5th pct	MC 95th pct	Pr(NPV < 0)	Mean reactivations
Conservative	5%	$66,866.14	$68,469	$851	1.88	$67,714	$24,277	$113,356	0.3%	66.8
	15%	$46,464.54	$47,890	$757	1.88	$47,201	$8,517	$87,927	2.4%	66.8
	30%	$15,862.13	$17,021	$616	1.88	$16,435	−$15,070	$49,499	18.1%	66.8
Base	5%	$241,533.35	$243,113	$1,152	1.37	$242,669	$184,510	$303,489	0.0%	128.8
	15%	**$201,501.66**	**$202,912**	**$1,025**	**1.38**	**$202,618**	**$150,709**	**$256,482**	**0.0%**	**128.8**
	30%	$141,454.13	$142,612	$835	1.39	$142,500	$99,966	$186,059	0.0%	128.8
Optimistic	5%	$464,630.73	$465,794	$1,403	0.83	$465,537	$395,658	$541,226	0.0%	206.1
	15%	$399,568.03	$400,610	$1,249	0.83	$400,458	$338,091	$467,804	0.0%	206.1
	30%	$301,973.98	$302,834	$1,018	0.85	$302,625	$251,721	$357,269	0.0%	206.1

ANALYTICAL EXPECTED NPV BY CELL ($ THOUSANDS)

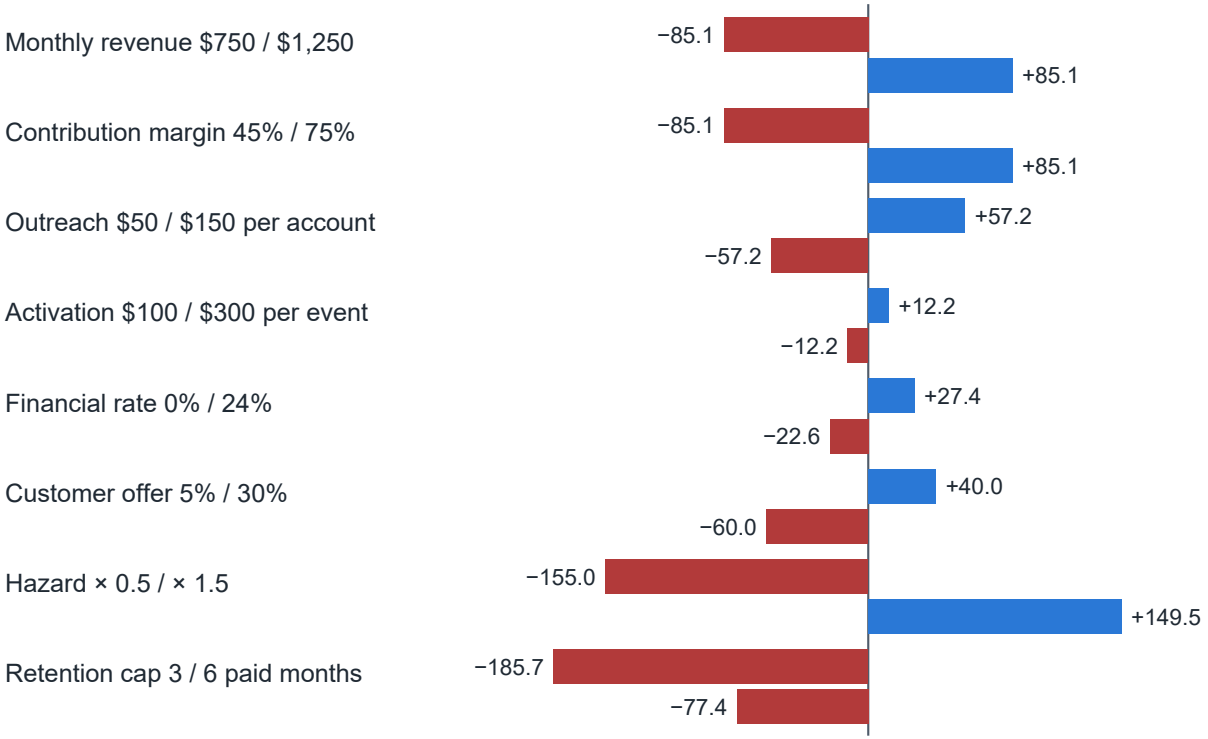


Read the table across one row at a time. Base / 15% is the reference cell used for routing priorities; the nine cells are alternative assumption sets and must not be summed. Every cell passes the predeclared |z| ≤ 4.5 reconciliation between simulation and closed form (maximum 1.88). Positive values in all 1,000 base-case runs do not establish zero real-world risk: parameters are fixed and root dependence is unestimated.

Analytical expected NPV is the closed-form expectation under the same billing and discounting rules; the Monte Carlo columns are finite-sample statistics from 1,000 independent replications per cell (PCG64, seed 20261009, common random numbers across cells). P5/P95 are conditional outcome percentiles, not confidence intervals for real program value.

Appendix E — Financial sensitivity, break-even and adverse stress

CHANGE IN EXPECTED NPV VS. BASE ($ THOUSANDS) — FIRST / SECOND VALUE



Blue = increase, red = decrease. Within each pair the upper bar is the first value named (e.g., $750), the lower bar the second (e.g., $1,250).

Root dependence: 96.8% of base / 15% expected NPV comes from the 257 of 295 root groups with more than one modeled account; parent-child billable independence is unverified and outcomes inside a root are modeled as independent. **Discount response:** no elasticity is estimated; the hazard multiples are hypothetical solutions.

ZERO-NPV BREAK-EVEN (BASE HAZARD, 5% OFFER)

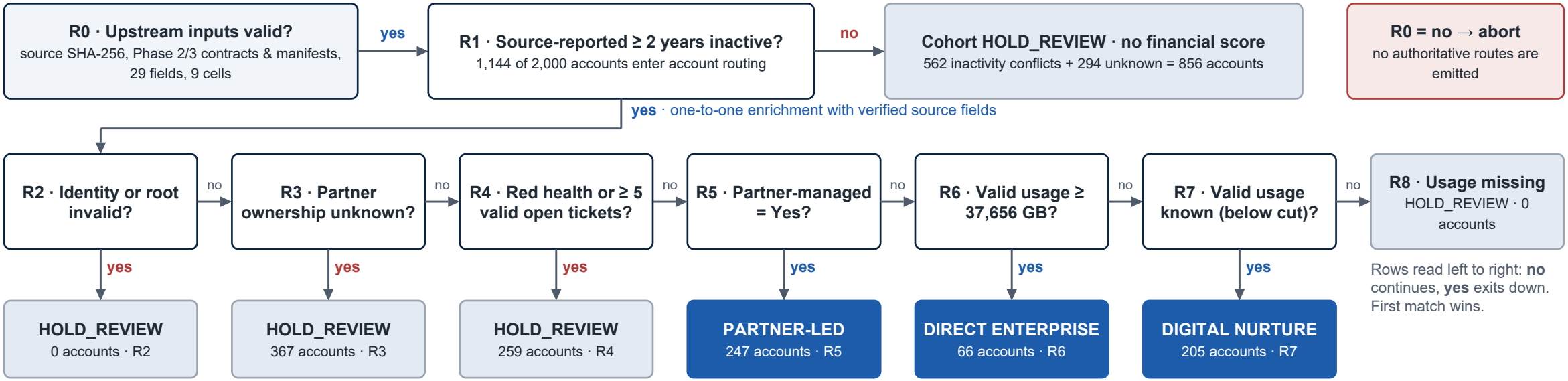
Minimum monthly gross revenue	**$364.89**
Minimum contribution margin	**21.89%**
Maximum outreach per account	**$311.13**
Maximum activation per event	**$2,182.41**
Hazard multiple for 15% to match 5%	**1.132×**
Hazard multiple for 30% to match 5%	**1.412×**

COMBINED ADVERSE STRESS — TWO DISTINCT VECTORS

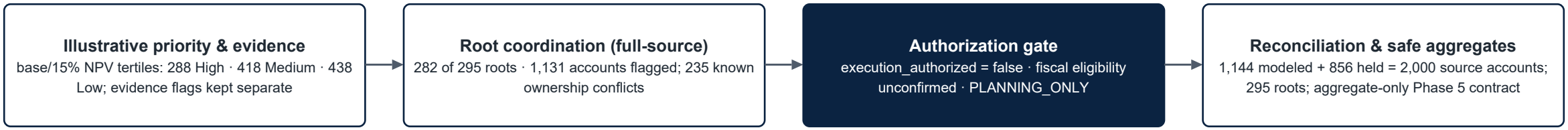
Assumption	Committed Phase 3	Alternative compounded
Hazard multiplier	× 1.0	× 0.5
Monthly gross revenue	$750	$750
Contribution margin	45%	60%
Outreach per account	$150	$150
Activation per reactivation	$300	$200
Annual financial discount rate	24%	12%
Maximum paid months	3	3
Hypothetical commercial offer	15%	15%
Analytical expected NPV	**−$124,647.43**	**−$124,704.05**

One-at-a-time changes around base hazard / 15% offer (analytical expected NPV $201,501.66). The two stress vectors are distinct assumption sets and are both reproduced exactly from frozen account hazards; neither replaces the primary nine-cell results.

Appendix F — Algorithmic GTM routing flowchart (policy v2)



APPLIED TO EVERY ROUTED ACCOUNT, AFTER THE RULES — NEVER OVERRIDING THEM



Totals: partner-led 247 · direct-enterprise 66 · digital-nurture 205 · hold/review 626 (R3 367 + R4 259) = 1,144 modeled accounts. Missing health or tickets add review flags without erasing an otherwise interpretable channel; unknown ownership always blocks channel assignment; a positive NPV never overrides a rule.

Rule order is exactly that of gtm_routing.route_account (policy phase4-routing-policy-v2). Counts are the frozen Phase 4 results. The 37,656 GB cut is the observed 75th percentile of the cohort — a scale heuristic, not verified enterprise status. Ticket counts are a snapshot backlog proxy; ticket velocity and revenue decay are unobserved.

Appendix G — GTM decision table, evidence quality and root coordination

RULE PRECEDENCE (FIRST MATCH WINS)

Rule	Condition	Disposition	Accounts
R0	Upstream source, contract, manifest or join fails	`Abort`	—
R1	Outside the source-reported two-year cohort	`Cohort hold, no score`	856
R2	Invalid identity or root relation	`HOLD_REVIEW`	0
R3	Partner ownership unknown	`HOLD_REVIEW`	367
R4	Red health or ≥ 5 valid open tickets	`HOLD_REVIEW`	259
R5	Known partner-managed = Yes	`PARTNER_LED`	247
R6	Known No and valid usage ≥ 37,656 GB	`DIRECT_ENTERPRISE`	66
R7	Known No and valid usage below the cut	`DIGITAL_NURTURE`	205
R8	Remaining case, including missing usage	`HOLD_REVIEW`	0

SERVICE SNAPSHOT AND REVIEW FLAGS (ACCOUNTS)

Critical backlog (≥ 5)	**189**	Partner review	**367**
Elevated (3–4)	**211**	Health review	**703**
Low (1–2)	**372**	Ticket review	**591**
Observed zero	**181**	Service recovery	**381**
Unknown / 999	**191**	Root coordination	**1,131**

ILLUSTRATIVE PRIORITY × EVIDENCE QUALITY

Priority	Critical	Material review	Relatively complete	Total
High	67	211	10	**288**
Medium	181	223	14	**418**
Low	249	167	22	**438**
Total	**497**	**601**	**46**	**1,144**

High-priority missingness: health unknown 187 of 288; partner unknown 23; 999-ticket sentinel 53. Unknown health is scored at the reference level, so financial order and evidence readiness must be read separately.

ROOT COORDINATION (295 MODELED ROOTS)

Multi-account modeled roots	**257**
Known Yes/No ownership conflict — modeled members	**145**
Known Yes/No ownership conflict — full-source members	**235**
Unknown-ownership ambiguity (full source)	**260**
Mixed planning channels inside the root	**215**
Coordination required (multi-account or full-source conflict)	**282**

0 authorized customer-contact actions. No campaign, partner notification or CRM update occurred; rollout needs eligibility, permissions and human review.

All counts are frozen Phase 4 aggregates (denominator 1,144 modeled accounts or 295 modeled roots). Evidence, service and priority classifications are planning diagnostics, not execution readiness; open tickets are a snapshot backlog proxy and ticket velocity is unobserved.

Appendix H — Validation evidence, limitations and next steps

VALIDATION EVIDENCE

Phase 1 data foundation	`1f0fa17`
Phase 2 survival	`1d8b136`
Phase 3 Monte Carlo NPV	`e94f174 + 3459169`
Phase 4 GTM routing	`38b1f43`

- Each phase was independently audited against a raw-CSV oracle and frozen before the next.
- Phase 5 build gate re-ran the source integrity check and the read-only Phase 4 semantic validator, pinned the input contract and manifest bytes, hash-checked every aggregate and re-derived both stress vectors before drawing any page.
- Frozen baseline test suite: 96 passed with warnings as errors in a disposable copy (2026-10-11); Ruff clean at the Phase 4 freeze.
- Every number on pages 1–13 is generated from those validated inputs; none is typed by hand.

PRINCIPAL LIMITATIONS

- No observed reactivation histories, campaign control group or revenue-decay series.
- Reactivation hazards, unit economics and retention are analyst assumptions.
- No price-response elasticity: offer comparisons hold hazard fixed.
- Incremental uplift is not causally identified (no untreated counterfactual).
- Parent-child billable independence and within-root dependence are unverified.
- Missing health and 999 tickets shape rankings through reference-level scoring.
- Tickets are a snapshot backlog, not velocity; fiscal eligibility is unconfirmed.
- Historical recorded spend is not ARR or recoverable revenue.

DATA NEEDED TO CALIBRATE, THEN NEXT STEPS

- Fiscal-period bookings by account to confirm dormancy and measure decay.
- Authoritative partner ownership and contact permissions by account and root.
- Current SFDC health and dated ticket history (velocity, not snapshots).
- Pilot outcomes with randomized holdouts: reactivation, uplift, cost, margin.
- Offer-level response data to estimate discount elasticity.

Next 3 actions

- Run the Gate 1 validation sprint on the 1,144 provisional accounts.
- Resolve ownership for the 282 coordination roots before any channel motion.
- Design a randomized pilot per channel; refit hazards and economics on its results.

Implementation readiness (code, controls, reproducibility) is established; operational readiness (eligible, permissioned, measurable outreach) is not. The model should be recalibrated on observed pilot outcomes before any financial claim is made.